

Applying and Verifying a Small Language Model on Inexpensive Hardware

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Abstract

We applied the Phi3 Mini SLM on a Raspberry Pi 5 in two domains - language translation and robot control. To compensate for hallucinations, we included a feedback loop in both applications. Although Phi3 Mini runs slowly, it suffices to test these proof-of-concept applications. In testing our applications, we found that the feedback loop enabled users to compensate to some extent for the limitations of the SLM.

Two drawbacks of generative AIs are that they are resource-hungry and prone to “hallucinations”. To address resource consumption, we created two applications that run on an inexpensive low-power device (the Raspberry Pi 5) and employ the Phi3 Mini Small Language Model (SLM) (Abdin et al. 2024). To address hallucinations, we compensate for errors in the SLM output with mechanisms for verification and user feedback. One application translates text between different languages. The other enables the user to command a robot using natural language. We restricted ourselves to customizing the behavior of the SLM solely through prompt engineering - we did not investigate fine-tuning.

The language translator has two feedback mechanisms. First, it gives the BLEU score (Papineni et al. 2002) for each translated output. Second, the user can request a translation back to the source language. If either mechanism suggests an unsatisfactory translation, the user gives the SLM extra instructions to improve the translation.

The robot controller uses the SLM to generate commands for an HTN planner (Nau 2013) that in turn controls the robot. The system first verifies the structure and components of the command. It then obtains user verification by providing a natural-language representation of the command given to the planner. If the command is incorrect, the user provides additional instructions via the SLM to clarify intent.

We learned from testing both systems with volunteers:

- A Raspberry Pi 5 runs the Phi3 SLM fairly slowly, but fast enough for our applications to function.
- We encountered significant scalability problems. Language translation was limited to about a paragraph, and for robot navigation its ability to distinguish locations decreased significantly as the number of locations increased.

- Feedback provided to the user increased the usability of both applications.

Language Translator

Language translator users specify the target language and the text to translate. The system sends this text combined with the prompts below to the SLM. These messages are intended to ensure that the system is primed for the user input and that the output consists of only the translated text.

- “You are an expert translator who translates text that the user gives you into a language of their choosing.”
- “When translating text, DO NOT add any additional comments, explanations, or pronunciations. This includes quotes explanations of the translations, parentheticals, etc. Just give the translated text.”
- “Unless prompted otherwise, do not give multiple translations for a text. Just give one translation.”

Users receive feedback from BLEU scores and from requesting translations back to the source language. BLEU outputs a score from 0 to 1 based on the quality of the machine translation. A score around 0.3-0.4 is considered acceptable; anything beyond that approaches human quality.

Robot Controller

The robot controller system provides a natural language interface to a hierarchical task network (HTN) planner running on a Raspberry Pi 5 atop an iRobot Create3. The system is initialized with a map of the robot’s floor and a list of packages and their locations, a description of the locations and packages, and aliases for the locations. The map and list of packages are used to initialize the HTN planner, the description is provided as input to the SLM, and the aliases are used to verify destinations with the user. Importantly, the description contains the coordinates of each location on the map. The concept was inspired by (Ding et al. 2023).

The system prompts the SLM with the user’s input along with a system message that contains the description of the locations and packages and is prefixed by the following text:

Based on the following input, you are to identify a package and the destination the user would like the package to be delivered to. Output your findings in

the following format `**package** **destination**`, replacing `**package**` with the name of the package and `**destination**` with the name of the destination, as identified from the input.

The system looks at the first two parts of the SLM output, ignoring any extraneous explanations. It verifies that the first part is a known package in the system and that the second part is a known destination. If the verification fails, it asks the user to “Please clarify an item and destination” and prompts the SLM with the original prompt and the additional information. If the system verification succeeds, the system asks the user for verification. If the user confirms their intent, the system calls the HTN planner to deliver the identified package to the identified location. Otherwise, the system asks the user to “Please clarify an item and destination” and prompts the SLM with the original prompt and the additional information.

Experiments and Results

We recruited ten volunteers to assess each system. For both systems, each communication with the SLM could take anywhere from 30 seconds up to 10 minutes. We had hoped to leverage the Raspberry Pi 5’s GPU to accelerate performance, but from what we could determine Ollama does not yet have a back-end that is compatible with it.

Language Translator

Ten total volunteers tested the language translator. On average, each test involved 2-3 rounds of translation before the participant was satisfied. Here were some key findings:

- Nine out of 10 users requested at least one retranslation.
- The best overall translations were from English into Spanish and German.
- Six of the ten participants chose to translate to and from Spanish. These translations were mostly competent. However, after prompting to reduce the use of present and past participles, the translation quality improved significantly.
- The other four participants chose French, Italian, German, and Russian. When translating into Russian, the system threw one Spanish sentence into the output.
- The model would often translate a passage very literally, failing to retain the style and nuance of the original.
- The BLEU score was consistently an intuitive metric for the translation quality. While there were a few instances of translations with low BLEU scores being readable, all translations with high BLEU scores were of high quality.
- The first-round translations would often neglect to translate a sentence, or introduce a gratuitous output sentence.
- It occasionally balked at translating, either repeating the original text untranslated or describing how it would perform the translation without actually doing it.

Robot Planning

Of the ten tests, the robot successfully delivered to the desired location four times. On two of these occasions, the system correctly identified the package and destination on the

first attempt, and, on the other two successful deliveries, the system identified the correct package on the first attempt and the correct destination after a user correction.

The SLM effectively adhered to the system message to categorize the input in terms of a desired state. However, its ability to identify the correct package and destination and to output them in the proper format was much poorer. It was much better at identifying the package than the destination, which has many more options. Likewise, the system appeared to qualitatively perform better during testing in the development phase, which was carried out using much smaller maps of a small shape, apartment, or portion of the floor. Despite the relatively poor performance of the model in identifying the user’s desired destination, verification of the prompt was important to the relative number of successes. Additionally, the number of overall successful tests were doubled, from two to four, with just a single round of user verification.

Based on the small sample size, it is difficult to identify general trends among prompts that were successful. However, it does seem as if prompts that provide extraneous information are likely to be unsuccessful. Of the three prompts that provided extraneous information, two in the form of identifying the current location of the package and one by instructing the robot to make no additional stops, none resulted in a successful delivery. In fact, the extraneous information that is implicit to the design of the system seemed to be the greatest obstacle to its success. It often assumed the current location of the package was also its destination, rather than the user-requested destination. Perhaps the system would be more successful if this information were omitted. However, on other occasions the model produced a different destination for no apparent reason. The SLM has no underlying logic model to determine which information is more important to producing a correct response, so it simply weighs all the information it is given equally, instead of following some logical sequence.

Conclusions and Future Work

The goal of this study was to assess the utility of the Phi3-mini SLM on inexpensive hardware in two task domains: language translation and robot planning. Drawing inspiration from recent work demonstrating the utility of SLMs (Wang et al. 2024), we sought to assess SLM utility based on prompt engineering and incorporating user feedback.

We found that, at least for now, the Raspberry Pi 5 using Ollama was sufficient for assessing the system prototypes we developed, but it is not sufficiently fast for practical use. In terms of task competency, both systems exhibited competence with relatively small problems but struggled to scale to larger problems. Fortunately, it was possible to address hallucinations through user feedback.

Future work will focus on identifying either a more effective SLM setup or alternative inexpensive computers that will allow for more natural response times. From there, we would like to try to more precisely characterize aspects of tasks that lead to both success and trouble with the SLM.

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